Natural Language Processing Journal 7 (2024) 100067

Contents lists available at ScienceDirect

Natural Language Processing Journal

journal homepage: www.elsevier.com/locate/nlp

Ensemble learning with soft-prompted pretrained language models for fact checking

Shaoqin Huang [a], Yue Wang [a,*], Eugene Y.C. Wong [a], Lei Yu [b]

[a] Department of Supply Chain and Information Management, The Hang Seng University of Hong Kong, Hong Kong
[b] Danzhou People's Hospital, Hainan, China

ARTICLE INFO

Keywords:
Fact checking
Social media
Ensemble learning
Soft prompt
Language model

ABSTRACT

The infectious diseases, such as COVID-19 pandemic, has led to a surge of information on the internet, including misinformation, necessitating fact-checking tools. However, fact-checking infectious diseases related claims pose challenges due to informal claims versus formal evidence and the presence of multiple aspects in a claim. To address these issues, we propose a soft prompt-based ensemble learning framework for COVID-19 fact checking. To understand complex assertions in informal social media texts, we explore various soft prompt structures to take advantage of the T5 language model, and ensemble these prompt structures together. Soft prompts offer flexibility and better generalization compared to hard prompts. The ensemble model captures linguistic cues and contextual information in COVID-19-related data, and thus enhances generalization to new claims. Experimental results demonstrate that prompt-based ensemble learning improves fact-checking accuracy and provides a promising approach to combat misinformation during the pandemic. In addition, the method also shows great zero-shot learning capability and thus can be applied to various fact checking problems.

1. Introduction

Infectious diseases, such as COVID-19, have caused unprecedented challenges to society. It has elicited a global response from the online community, resulting in a flood of new information about the virus's pathogenesis, new treatments for infected patients, and vaccines under development. These networks are known to quickly reach enormous personal networks (Chen et al., 2022). Much of the information available on the internet is valuable, but others, particularly misinformation and disinformation, can be extremely dangerous (Peng et al., 2023). False and misleading information may jeopardize efforts to control the pandemic. As a result, the flood of online information necessitates fact-checking of online resources and claims relevant to infectious diseases.

Fact-checking can be defined as a task of assessing the truthfulness of a claim based on evidences as shown in Fig. 1 (Vlachos and Riedel, 2014). Due to the increasing demand of fact-checking in different domains, many fact-checking organizations were founded, such as FactCheck, Full Fact, NewsGuard PolitiFact, and Snopes. Fact checkers usually undertake meticulous manual checks on supposed facts. Understanding the issue, identifying claims, collecting evidence, confirming assertions and their explanation, and checking for errors are among the tasks (Hanselowski et al., 2019). The speed and effectiveness of manual fact-checking, on the other hand, cannot keep up with the rate at which information moves throughout the Internet. Furthermore, verifying medical information such as COVID-19 necessitates extensive medical understanding of the claims' topic (Kotonya and Toni, 2020). In their investigation of COVID-19 misinformation subjected to manual scrutiny by fact-checkers, Brennen et al. discovered that fact-checking entities faced significant challenges in effectively managing the extensive volume of information disseminated through social media platforms (Brennen et al., 2020). As a result, there is a need for automatic fact-checking tools to validate infectious diseases-related claims or news in order to assist information searchers in evaluating material obtained from the Internet.

Recent advances in deep learning and natural language processing shed light on the automatic identification of misinformation distributed on the Internet, making the development of an automatic fact-checking tool promising. To counteract the spread of disinformation, researchers have examined assessing the credibility of a given claim against textual sources (e.g., scientific publications, Wikipedia) that may support, refute, or provide insufficient information to arrive at a conclusive verdict (Wadden et al., 2020).

However, challenges persist for the automatic fact-checking of infectious diseases related claims. Claims spread on the internet are usually informal and vague, consisting of idiomatic and ambiguous expressions, whereas the evidence used to check the claims is mostly from scientific

* Correspondence to: RM615, Block D, Siu Lek Yuen, Shatin, Hong Kong Special Administrative Region of China.
E-mail address: yuewang@hsu.edu.hk (Y. Wang).

https://doi.org/10.1016/j.nlp.2024.100067
Received 27 November 2023; Received in revised form 1 March 2024; Accepted 19 March 2024

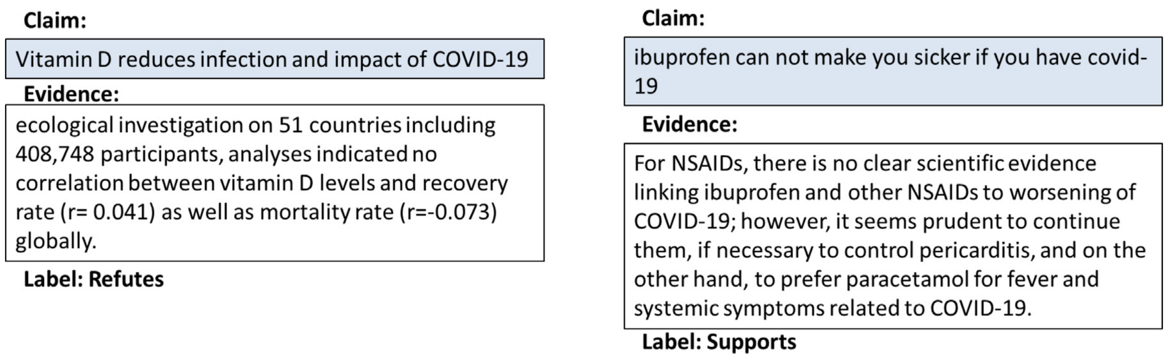

Fig. 1. Examples of evidence based fact checking.

articles. They are formal, exact, and accurate. Thus, a semantic gap exists between the claims in layman languages and the evidence in formal scientific languages. In addition, a real-world claim may contain multiple aspects. Some aspects may be correct, and some may be wrong (e.g. "People with diabetes and coronary heart disease have higher risk for COVID-19"). Thus, the fact-checking process can be complicated as the claim may be supported by certain evidence but refuted by others. For example, research indicates that people with diabetes have higher risk for COVID-19, but there is no correlation between coronary heart disease and COVID-19. Thus, the previous claim is only partially correct.

To meet these challenges, we proposed a soft prompt base framework to tackle the infectious diseases fact checking problem using COVID-19 as an example. A prompt refers to the input text or instruction given to a pretrained language model (PLM) to generate desired outputs or responses (Liu et al., 2023). Unlike hard prompts, which provide a hand-crafted guideline context for language models, a soft prompt does not need the manually created guideline. It provides a loosely defined or less constraining input, enabling the model to explore various creative solutions or generate diverse outputs (Liu et al., 2023). Thus, soft prompts can possess better generalization capability in natural language process tasks. We also explore various soft prompt structures and ensemble them to better understand complex material in short, informal, and unstructured texts from social media. The ensemble model could capture various linguistic syntactic cues, semantic features, and contextual information that arise in COVID-19-related text data. This ensemble approach enables the model to efficiently generalize new claims by combining the collective knowledge and strengths of individual classifiers.

To summarize, we make the following contributions in this paper. First, a prompt-based network structure is developed for the COVID-19 fact checking task. To combat the issue of the language model lacking explicit guidance in soft prompt, we develop various soft prompts to solve the ambiguity and uncertainty in soft prompt-based text classification. Second, we ensemble different soft prompt structures to fully leverage the language cues and syntactics and ensemble semantic information extracted by individual soft prompts. Third, state-of-the-art performance is achieved on three data sets on COVID-19 fact checking. In addition, experiments demonstrate that the proposed methods have great zero-shot capability and can thus be applied to other fact checking issues.

2. Relevant work

Automated fact-checking has received increasing attention in recent years. Researchers developed fact-checking methods with or without external evidence. Fact-checking with external evidence requires the integration of information from other textual or structural sources into the fact-checking process, whereas the latter relies on linguistic patterns to identify misinformation.

2.1. Fact checking without evidence

Research on fact checking without evidence usually adopts stylometric approaches, i.e., to extract stylistic linguistic features from text. It was observed that misinformation may contain certain idiosyncrasies such as unique styles or language characteristics. This stream of method attempts to identify the idiosyncrasies and determine whether or not the claim is reliable. Enos et al. (2007) utilized decision trees to identify critical segments in a text and use the segments to detect deceptive statements. Mihalcea and Strapparava (2009) identified dominant word classes in deceptive text using LIWC. They used Naïve Bayes and SVM to determine whether the statement regarding certain topics were true or not. Similarly, Ott et al. (2011) used unigram, bigram, and trigram features of text and combined the features with LIWC features to train the deceptive content detector based on Naïve Bayes and SVM. Feng et al. (2012) derived features from Context Free Grammar (CFG) parse trees. They used this experiment to demonstrate that these features were better than the previous lexico-syntactic features in deception detection. Afroz et al. (2012) extracted lexical features, syntactic features, content specific features, and lying detection features from text and fed them into SVM to detect stylistic deception in text.

Recently, pretrained language models (PLM), such as BERT and GPT, have revolutionized the research in NLP. The PLMs were trained using a massive amount of text and have been acknowledged to contain knowledge in various domains. Researchers also leverage this kind of implicit knowledge in PLM in fact-checking tasks. Lee et al. (2020) constructed evidence text by unmasking the entities of the original claim using BERT. The evidence from the language model is then transmitted to the entailment model, which predicts whether the claim is supported or refuted.

Fact-checking without evidence has limitations. Stylometric approaches fail to distinguish well-presented misinformation that does not have stylistic linguistic misinformation features. When a PLM is used to check the claims, the models are likely to propagate the biases in the training date (Guo et al., 2022). Furthermore, these models are biased to seek for and extract knowledge from the model (Sung et al., 2021).

2.2. Fact checking based on external evidence

Fact checking based on external evidence enables the system to make informed decisions based on external sources, thus overcoming the limits of fact checking just based on claims themselves. Furthermore, evidence can provide justifications for fact-checking verdicts to end users (Sarrouti et al., 2021). Due to this reason, evidence-based fact checking has been prevailing.

In evidence-based fact checking, researchers leverage different external resources to identify misinformation in text. Fact checking using external evidence has been prevailing in the research community. The external evidence used can be knowledge bases that provide structured information in relevant domains. Vlachos and Riedel (2015) developed a distantly supervised method using Freebase to verify simple statistical claims. As a result, the probability that a claim is reliable can be calculated. Ciampaglia et al. (2015) showed that manual fact checking can be modeled as finding the shortest path between nodes on knowledge graphs. They deployed a public knowledge graph extracted from Wikipedia to verify the claims in the areas of entertainment, history, and others. This enables calculating the likelihood of a claim being true using a knowledge network as a topology.

The majority of the external resources used are unstructured online resources such as websites and news articles. Popat et al. (2018) retrieved supporting online articles for or against allegations to mimic the traditional news fact-checking procedure. They also developed a model to assess the credibility of the claim source. Ferreira and Vlachos (2016) employ news headlines as evidence to verify claims in a related work on attitude detection. Other methods use article abstracts rather than headlines or titles as evidence for fact checking (Alhindi et al., 2018; Hanselowski et al., 2019). These approaches usually need to delete irrelevant sentences to find fine-grained substantial evidence. Similarly, Wadden et al. (2021) validate scientific assertions using evidence from research papers' abstracts. Pair-level supervised contrastive learning (PairSCL) adopts a cross attention module which can identify the joint representations of the claim-evidence pairs (Li et al., 2022).

Fact checking has also been studied in healthcare related domains. There is growing interest in utilizing biomedical texts as evidence for fact-checking health claims. For instance, Arana-Catania et al. (2022) compiled a dataset of heterogeneous COVID-19 claims and explored various natural language inference methods to assess claim veracity. Instead of employing human annotators, Saakyan et al. (2021) constructed a medical claim dataset containing truthful claims, source articles, and counterclaims. They also establish a baseline for claim verification using RoBERTa. Wührl and Klinger (2022) leveraged online user-generated health content to extract entity-based claim representations. This can improve the effectiveness of multiple fact-checking models. Some research utilizes biomedical corpora to generate human-comprehensible rationales for fact checking. For instance, Kotonya and Toni (2020) employed extractive and abstractive summarization techniques to construct explanations of the fact checking result. They discovered that simultaneously optimizing a fact-checking model and an explanation generation model using domain-specific data could improve the checking performance of health claims.

The PLM has also proved to perform well in various fact checking datasets. Researchers have investigated whether PLM allows for correct fact checking in verb-complement constructions (Ross and Pavlick, 2019). It was observed that PLM tends to assume that verbs are veridical in text. PLM inferences are sensitive to different forms of verbs, such as negation and complement clauses. In addition, PLMs were trained using general domain corpora. To improve PLMs' performance in domain specific fact checking tasks, researchers usually combine external knowledge bases with PLM. Typical methods include supplementing lexical retrieval with dependency relations, using domain specific PLMs (such as BioBERT and SCIBERT), and including domain specific knowledge or features. Experiments indicate that if the domain specific features are added to BERT, the efficiency of fact checking model training can be significantly improved (Lin and Su, 2021). This shows that domain knowledge is extremely beneficial. However, Sushil et al. (2021) noticed that the inclusion of external domain knowledge in the neural network may lead to unreliable knowledge augmentation. Thus, it is necessary to develop effective ways of incorporating knowledge to the model. To summarize, PLM based methods, as compared to older methods, enhance text representation, achieve greater generalization capability of the model, and enable more sophisticated calculations without relying on hand-crafted rules (Talman and Chatzikyriakidis, 2019).

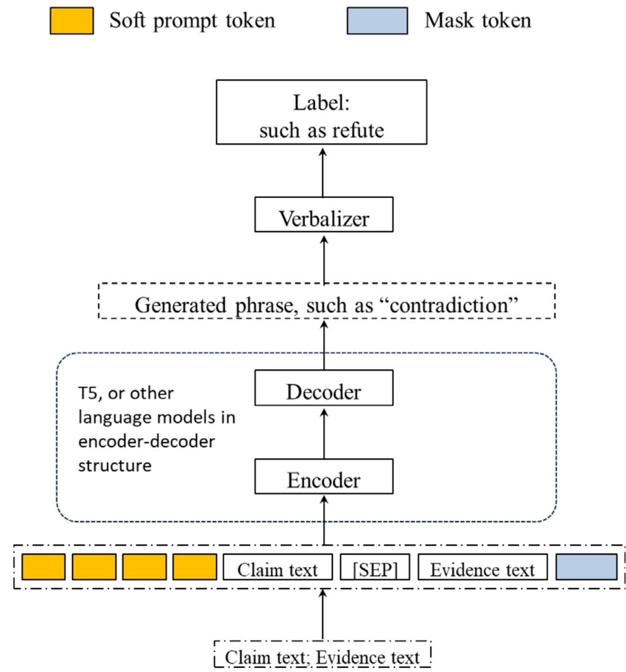

Fig. 2. Soft prompt based fact checking.

3. Research methodology

3.1. Overview

The recently developed prompt-based methods have demonstrated outstanding performance in NLP tasks by better exploiting PLM (Liu et al., 2023). Prompts can enhance the efficiency of the language model in adapting general information to a particular domain (the COVID-19 domain in this paper) due to the similar text format, syntax, and semantics. Thus, the prompt-based model does not necessitate a massive amount of additional data to fine-tune the language model. Due to this reason, we deploy a prompt-based framework for COVID-19 fact checking tasks. This study examines fact checking from a text classification perspective. Given a claim Q, and a set of evidence E, the objective is to verify the factuality of the claim with respect to the evidence set E with labels support, refute, and neutral or NEI, which represent there being not enough information and an inability to make any decision.

3.2. Methodology

3.2.1. Soft prompt-based fact checking

The language model of T5 is used in this paper (Raffel et al., 2020). Similar to BERT (Devlin et al., 2019) and GPT (Brown et al., 2020), T5 is also based on a Transformer unit but consists of both encoder (like BERT) and decoder (like GPT) in the model (Khan et al., 2023). This encoder–decoder structure is advantageous due to its ability to better capture contextual information, handle variable-length input and output sequences, and enable conditional generation. In the encoder–decoder structure of T5, the fact-checking is solved in a text-to-text format as shown in Fig. 2.

The prompt-based method functions to reformulate the downstream tasks as a cloze problem so that the objective of the downstream task is modeled the same as the original training procedure of PLM. Let $\mathbf{x} = \{x_1, x_2, \ldots\}$ be a claim text where x_i is the ith word in the claim and $\mathbf{y} = \{y_1, y_2, \ldots\}$ be a evidence text where y_j is the jth word in the evidence. In fact-checking tasks, the original input of claim $\mathbf{x}$ and evidence $\mathbf{y}$ can be reformulated to the format of “[CLS]$\mathbf{x}$[SEP]$\mathbf{y}$[SEP] [Prompt]

[MASK]" where [CLS] appears at the beginning of the text and is a special classification token. The [SEP] token separates different text in the whole input. It also indicates the ending of the input. [Prompt] and [MASK] form a complete sentence to indicate the relationship between the claim-evidence data pair **x** and **y**. Thus, the corresponding [MASK] may be support, refute, or neutral. The text structure in this new formulation is, in fact, a cloze question where PLM should fill in the [MASK] blank, which is exactly how PLM is pretrained and brings better performance in class label prediction.

When using a prompt-based method, a hard prompt, which is designed by humans (Schick and Schütze, 2021), is needed to describe the downstream task in a human-interpretable way. For example, if we want to determine the relationship between two sentences – "It is wet outside" and "It is rainy today" – we can construct a new text: "[CLS] It is wet outside [SEP] It is rainy today [SEP] What is the relation between the two sentences? [MASK]" and feed it into a PLM. "What is the relation between the two sentences?" is a human designed prompt to guide the PLM to produce the label "support" at the end. However, hard prompts are human crafted. When facing a new task, new hard prompts need to be designed. Furthermore, it is not clear how to phrase a good prompt.

In this paper, we devise a soft prompt structure for fact checking. It is noticed that human-interpretability of prompt is unnecessary for MLM to understand the downstream task (Li and Liang, 2021). Therefore, we used a series of parameterized tokens trainable in the continuous embedding space of language models as the soft prompt and human generated prompt is not needed. For example, when using soft prompt, the previous fact-checking task is reformulated as "[CLS] It is wet outside [SEP] It is rainy today [SEP][UNUSED1][UNUSED2][UNUSED3][MASK]" where each [UNUSED] token is a randomly initialized vector with a dimension of embedding layer in PLM. Thus, no human-generated prompt is needed. After feeding the reformulated text to PLM, soft prompt can be a powerful trigger to predict the label of the claim-evidence pair. To further incorporate domain knowledge related to COVID-19 to the PLM, we further fine tune the PLM to improve the fact checking results. In addition, we also investigate different soft prompt structures and identify the one with the best performance to check the claims.

For a claim text $\mathbf{x} = \{x_1, x_2, \dots\}$ and an evidence text $\mathbf{y} = \{y_1, y_2, \dots\}$, we transform them into a new sequence $\mathbf{x}_{prompt}$ =[CLS][UNUSED1]... [UNUSED-k] x [SEP] y [SEP] [MASK] so that the new sequence has the format of the T5's input. Then the fact checking can be converted to a text-to-text setting by generating a verbalizer word $v(\mathbf{x}_{prompt})$ using T5 as shown in Fig. 2. A verbalizer word indicating the relationship between the claim and evidence will be generated.

It should be noted that unlike previous research that mainly manually defines a set of verbalizer words indicating the support, refute, and neutral relationship such as {support, refute, neutral, entail, contradiction, NEI, undetermined, true, false, favor, against}, we set no constraints on the choices of verbalizer $v(\mathbf{x}_{prompt})$. The reason is that we noticed that the relationship of weak support and weak refute between claim and evidence can hardly be described by the abovementioned predefined verbalizers, even with a more refined verbalizer selection, such as weak support, strong support, weak refute, and strong refute. The weak support and refute relationships are often misidentified as a neutral relationship. To solve this issue, we do not pre-define a set of verbalizers but just let T5 generate the verbalizer $v(\mathbf{x}_{prompt})$ based on the input, i.e., $\mathbf{x}_{prompt}$ without any constraints. This mechanism allows for more expressive and flexible outputs compared to an encoder structure like BERT, where the text encoding is mapped to pre-defined labels. Specifically, given the claim and evidence text, the possible relationship between them is determined by $P(v|\mathbf{x}, \mathbf{y})$ which is calculated by T5 decoder where v is the verbalizer. Then, we calculate the distance between the generated verbalizer's embedding with the embeddings of the three class labels, i.e., support, refute, and neutral. The fact checking result is the label which has the shortest distance with the generated verbalizer. Cosine distance is used in this paper due to its great performance and simplicity. The fact checking result is $\underset{i \in support, refute, neutral}{\operatorname{argmin}} \frac{h_v \cdot h_i}{|h_v| \cdot |h_i|}$ where h_v (h_i) is the embedding vector of the verbalizer word (the ith label).

The soft prompt is not manually crafted but optimized during the model training stage. During the model training stage, the encoding of the optimal soft prompt is found by minimizing the cross-entropy loss $-\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{3}\log P(z_k|\mathbf{x}_i, \mathbf{y}_j)$ where N is the total number of claim-evidence pair and z is the label (relationship, i.e., support, refute, and neutral in the fact checking task) of claim $\mathbf{x}_i$ and evidence $\mathbf{y}_j$. The procedure of prompt tuning for fact checking can be found in Fig. 2.

3.2.2. Ensemble of the soft prompt-based method

The soft prompt-based method uses certain numbers of [UNUSED] tokens in the prompt template to avoid the manual work of hand crafting the hard prompt. However, there is no systematic study to indicate how to set the optimal number of [UNUSED] tokens and how to place them in the template. In this paper, we explored different formats of soft prompt templates, including:

$\mathbf{x}_{prompt(k,0)}$ =[CLS] [UNUSED1]... [UNUSED-k] **x** [SEP] **y** [MASK] [SEP] with k being 4, 6, and 8;

$\mathbf{x}_{prompt(0,k)}$ =[CLS] **x** [SEP] **y** [UNUSED1]... [UNUSED-k] [MASK] [SEP] with k being 4, 6, and 8; and

$\mathbf{x}_{prompt(k,k)}$ =[CLS] [UNUSED1]... [UNUSED-k] **x** [SEP] **y** [UNUSED1] ... [UNUSED-k] [MASK] [SEP] with k being 2, 3, and 4.

It should be noted that different soft template structures could elicit different contextual and semantic information in the text, and thus yield different performance, which has been observed in our experiment. Thus, we ensembled all the classifiers to fully leverage the information extracted using different soft prompts (Mienye and Sun, 2022). Model ensemble refers to the technique of combining outputs from multiple models to improve overall performance. In our case, we train multiple T5 classifiers with different soft prompts, which are specific instructions given to the model during training to guide its fact checking process. These soft prompts allow each model to capture different aspects of the text and potentially uncover diverse patterns or insights. By combining the predictions from these models, we can leverage their collective knowledge and diversity to achieve more robust and accurate classifications.

We used hard voting to ensemble all the models. Hard voting is a method where each model in the ensemble independently predicts the class label for a given input text (Dong et al., 2020; Luo et al., 2023). These individual predictions are then combined by taking a majority vote. In other words, the class label that receives the most votes from the models is selected as the final prediction. In our case, each of the twelve soft-prompt based classifiers provide a predicted label, i.e., support, refute, and neutral. The majority of the twelve labels is used as the final fact checking label. The process can be found in Fig. 3.

4. Experiments settings

4.1. Dataset

The proposed method was assessed on three datasets related to COVID-19 fact checking, i.e., HealthVer, COVIDFact, and CoVERT. The details of the three datasets are as follows.

HealthVer (Sarrouti et al., 2021): The HealthVer dataset comprises 14,330 COVID-related claims extracted from evidence sentences retrieved by a search engine in response to 80 common questions on COVID-19. The claims were verified against abstracts from the CORD-19 dataset (Wang et al., 2020b). Claims in HealthVer can be complex, and the dataset includes naturally occurring refuted claims from the evidence sentences. For each claim, HealthVer provides candidate abstracts from the CORD-19 dataset as evidence.

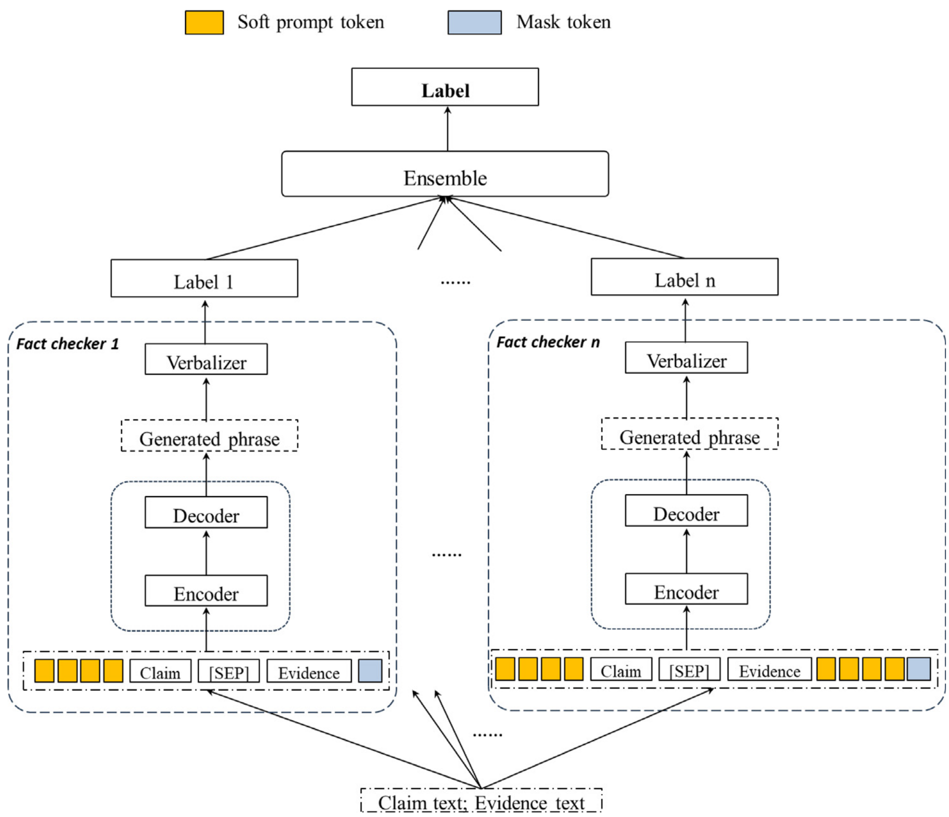

Fig. 3. Ensemble of soft prompts-based fact checking.

COVIDFact (Saakyan et al., 2021): The COVIDFact dataset is structured similarly to HealthVer but with only two labels of support and refute. COVIDFact gathers 4,096 COVID-19 claims from a subreddit and checks them against relevant scientific papers and news articles found through Google search. The dataset matches each claim to pertinent evidence sentences that were manually labeled as either supporting or contradicting the claim.

CoVERT (Mohr et al., 2022): The CoVERT dataset contains 300 tweets with real-world biomedical claims pertaining to COVID-19. Crowdworkers then investigated each tweet to assign a fact-checking label (support, refute and NEI) and evidence from the web to justify their verdict.

4.2. Performance measure and implementation details

Since COVID-19 fact checking is treated as a classification task, we use prevailing metrics of precision, recall, and F1 as the performance measure (Wang et al., 2020a; Zhuang et al., 2023). We use T5-small as the backbone language model to implement the soft prompt-based ensemble approach. An Adam optimizer is used with a learning rate of 3e−5. The maximum length of sequence and batch size are 512 and 16. For each of the three datasets, we randomly partitioned the whole dataset in the proportion of 80%:10%:10% to get the training, validation, and testing set.

Popular text classification methods are used as the baselines, including fastText (Bojanowski et al., 2016), CNN (Luo et al., 2022), BiLSTM (Wang et al., 2022), and BERT (Devlin et al., 2019; Luo and Wang, 2019), among which BERT and its variants have achieved state-of-the-art performance in previous work. In CNN, the filter sizes are 3, 4, and 5 with dropout rate being 0.5. The learning rate is 32 for CNN, BiLSTM, and BERT.

5. Experiment results

5.1. General findings

The performance of the proposed approach and the other baselines on the three datasets are shown in Table 1. For HealthVer, the proposed approach could yield the best performance, with the overall macro-F1 value being 0.7901, achieving the state-of-the-art performance. We noticed that the Transformer based structure, such as BERT and T5, generally achieve much better performance than other deep learning units such as CNN and LSTM. This is due to the outstanding capability of extracting semantic features from text (Wang et al., 2021).

On the dataset of COVIDFact, the advantage of the soft prompt-based method is even more significant. It should be noted that the data quality of COVIDFact is relatively worse than the other two due to the generation of data in the refute class. To produce more refuted claims, COVIDFact automatically substitutes some salient words in the original claims with a negation term. However, manual inspection found that this process yielded truly refuted claims only about one-third of the time. The remaining instances led to ungrammatical claims or claims where the provided evidence was irrelevant. However, the proposed method significantly outperforms other baselines. This shows the effectiveness of the prompt structure and ensemble framework.

None of the methods' performances on CoVERT are as good as HealthVer. The reason is that CoVERT is a relatively small dataset, only containing 300 tweets with real-world biomedical claims pertaining to COVID-19. In addition, the claims in the refute class of CoVERT are tweets structured in more free form than the other two, containing many slangs. The format of claims and evidence are significantly different. As a result, previous work also achieved unsatisfactory results (Mohr et al., 2022). Furthermore, we noticed that the class imbalance also has an effect on the overall performance. For example, in CoVERT,

Table 1
Performance comparison of different methods on the three datasets.

Method	HealthVer			COVIDFact			CoVERT		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
fastText	0.6444	0.6493	0.6468	0.5746	0.5592	0.5653	0.4437	0.3875	0.4024
CNN	0.6451	0.6389	0.6420	0.5287	0.4828	0.5047	0.3983	0.3778	0.3855
BiLSTM	0.5923	0.5978	0.5951	0.5616	0.5233	0.5283	0.4365	0.4175	0.4226
BERT	0.7315	0.7252	0.7283	0.4566	0.6758	0.5451	0.6822	0.5269	0.5420
Soft prompt with ensemble	0.7983	0.7862	0.7901	0.7775	0.7743	0.7759	0.6580	0.6061	0.6265

Table 2
Performance of the model trained on HealthVer but tested on COVIDFact and CoVERT.

	COVIDFact (two labels)	CoVERT (three labels)
Precision	0.6395	0.6478
Recall	0.4332	0.5859
F1	0.4511	0.5984

the distribution of the three classes is 33:11:6, but in HealthVer, the distribution is more balanced. This also partially explains the relatively worse performance of the proposed method in CoVERT. With such issues of data quantity and quality, we can see that the proposed method also achieves the best results. It outperforms BERT, the second best method, by 15.6% in terms of F1 score.

It should be noted that T5 deploys an encoder–decoder structure and BERT uses an encoder structure. The encoder–decoder structure allows for more expressive and flexible outputs compared to an encoder structure like BERT, where the text encoding is mapped to pre-defined labels. In our approach, we utilize the encoder–decoder structure and employ a verbalizer to map all the generated outputs to the desired class labels. By training the model to generate text conditioned on the input, encoder–decoder models can better capture the nuances and intricacies of the input text, resulting in more accurate and context-aware classifications. These advantages make T5 more suitable for tasks that require more expressive and flexible outputs beyond simple classification labels.

5.2. The zero-shot capability of the proposed method

In the context of COVID-19 fact checking, where new claims and evidence emerge rapidly, the ability to generalize to unseen instances is crucial. Thus, it would be necessary to investigate the generalization capability of the proposed model. In this subsection, we would like to explore the zero-shot capability of the proposed method. Since HealthVer is the largest among the three, we trained the method using the HealthVer dataset and tested it on CoVERT and COVIDFact. The results can be found in Table 2.

It can be noted that the proposed method has a good generalization capability. The fact checking performance is slightly worse than the results in the previous section for the CoVERT dataset. The F1 value is 0.5984 for the zero-shot setting and 0.6265 for the tuned method using CoVERT. But the difference is not significant. The p-value of the two-sample p-test is 0.894 for accuracy in the test, showing an insignificant difference between the zero-shot and non zero-shot setting.

The great zero-shot capability is due to the prompt-based ensemble structure. By training multiple classifiers on a range of soft prompts, the ensemble model learns to capture various linguistic syntactic cues, semantic features, and contextual information that arise in COVID-19-related text data. This ensemble approach enables the model to generalize well to new claims by combining the collective knowledge and strengths of individual classifiers. The soft prompts provide a broad understanding of the domain, enabling the ensemble model to make informed predictions even on previously unseen claim-evidence pairs. The diversity and adaptability offered by ensembling soft prompts make it a powerful technique for enhancing zero-shot capability in COVID-19 fact-checking, ultimately contributing to more accurate and reliable identification of misinformation and supporting evidence in real-time scenarios.

However, we noticed that the result of the zero-shot setting is worse for the COVIDFact dataset. The F1 value of COVIDFact is 0.4511 for the zero-shot case, which is significantly lower than the tuned network, i.e., the non-zero shot case, which has F1 value of 0.7759 (p-value = 0.002 for the test on accuracy). The reason for the worse performance of the zero-shot experiment in COVIDFact is due to the difference in the data format. HealthVer consists of a claim-evidence data pair annotated with three labels. However, the data pairs only have two labels of support and refute in the COVIDFact dataset. In addition, as mentioned in the previous subsection, the data quality of COVIDFact is relatively worse in that much of the data in the refute class was mainly generated by simply substituting some salient words in the original claims with a negation term. Thus, the issues on data quality and labels setting make it hard for the classifier trained on HealthVer to correctly label the data in COVIDFact. CoVERT and HealthVer have much more commonality in terms of labels, data sources, and text format. Thus, the model trained on HealthVer can perform well on CoVERT.

5.3. Ablation study

5.3.1. Effects of model ensemble

We investigate the impact of model ensemble and soft prompts on performance. To demonstrate the effectiveness of model ensembling, we compare the performance of the proposed method with the results obtained using single soft prompts $\mathbf{x}_{prompt(4,0)}\mathbf{x}_{prompt(6,0)}\mathbf{x}_{prompt(8,0)}$ $\mathbf{x}_{prompt(0,4)}\mathbf{x}_{prompt(0,6)}\mathbf{x}_{prompt(0,8)}\mathbf{x}_{prompt(2,2)}\mathbf{x}_{prompt(3,3)}$ and $\mathbf{x}_{prompt(4,4)}$. The result is shown in Table 3.

The table shows that ensembling different soft prompts can improve the performance of fact checking. Among the soft prompt structures, $\mathbf{x}_{prompt(4,4)}$ =[CLS][UNUSED1]... [UNUSED-k][MASK]x[SEP] y[UNUSED1] ...[UNUSED-k] [SEP] can yield the best overall performance. By placing more [UNUSED] tokens before and after the [MASK] token, we can effectively increase the context window that the model can consider when making predictions. This extended context window allows the model to capture more information about the surrounding words and their relationship to the masked token. Thus, $\mathbf{x}_{prompt(4,4)}$ is generally better than other prompt structures, although cannot completely dominate other prompt structures' performance.

The reason is that ensemble methods allow for the combination of multiple classifiers which are trained on different soft prompts, resulting in a diverse set of predictions. By aggregating these predictions, the ensemble model can leverage the unique strengths and insights of each individual classifier. This leads to enhanced overall performance and generalization capability. Additionally, ensembling soft prompts helps mitigate the impact of biases or errors inherent in any single prompt by introducing a broader range of linguistic cues and semantic perspectives (Luo et al., 2023). This diversity of prompts promotes a more comprehensive understanding of the text data, capturing different nuances, contexts, and linguistic variations that may be present in the dataset. Furthermore, ensembling soft prompts helps to address the issue of overfitting, as different prompts encourage the model to learn more generalized representations of the text. This combination of strengths, diversity, and improved generalization makes ensembling an effective strategy for boosting the performance and robustness of text classification models.

Table 3
Performance comparison of the ensemble model and the soft-prompt based model.

Method	HealthVer			COVIDFact			CoVERT		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
$\mathbf{x}_{prompt(4,0)}$	0.7246	0.7621	0.7385	0.7193	0.7313	0.7252	0.6130	0.5511	0.5734
$\mathbf{x}_{prompt(6,0)}$	0.7212	0.7720	0.7401	0.7544	0.7490	0.7517	0.6241	0.6046	0.5821
$\mathbf{x}_{prompt(8,0)}$	0.7142	0.7695	0.7434	0.7553	0.7458	0.7505	0.6311	0.5687	0.5906
$\mathbf{x}_{prompt(0,4)}$	0.7141	0.7307	0.7279	0.7591	0.7181	0.7380	0.6167	0.5662	0.5825
$\mathbf{x}_{prompt(0,6)}$	0.7194	0.7589	0.7386	0.7423	0.7259	0.7340	0.6266	0.6064	0.6107
$\mathbf{x}_{prompt(0,8)}$	0.7231	0.7431	0.7371	0.7354	0.7350	0.7352	0.6342	0.5893	0.6082
$\mathbf{x}_{prompt(2,2)}$	0.7173	0.7408	0.7321	0.7477	0.7182	0.7326	0.6117	0.5700	0.5838
$\mathbf{x}_{prompt(3,3)}$	0.7317	0.7409	0.7369	0.7414	0.7219	0.7315	0.6516	0.5792	0.6068
$\mathbf{x}_{prompt(4,4)}$	0.7431	0.7498	0.7464	0.7479	0.7245	0.7360	0.6478	0.5859	0.5984
Ensemble	0.7983	0.7862	0.7901	0.7775	0.7743	0.7759	0.6580	0.6061	0.6265

Table 4
Performance comparison of the T5 and the best soft-prompt based model.

Dataset	Method	Precision	Recall	F1
HealthVer	T5	0.7294	0.7323	0.7309
	$\mathbf{x}_{prompt(4,4)}$	0.7431	0.7498	0.7464
COVIDFact	T5	0.7291	0.6933	0.7107
	$\mathbf{x}_{prompt(4,4)}$	0.7479	0.7245	0.7360
CoVERT	T5	0.6072	0.5219	0.5458
	$\mathbf{x}_{prompt(4,4)}$	0.6478	0.5859	0.5984

5.3.2. The effectiveness of soft prompt

We would also like to study whether soft prompts can significantly improve the fact checking result. To make a fair comparison, we show the fact checking result of the vanilla T5 model and the soft prompt $\mathbf{x}_{prompt(4,4)}$ based classification results, as the vanilla T5 cannot be ensembled.

Table 4 shows that the soft prompt-based structure yields better overall performance than the T5-based method. It can be noticed that the performance improvement is the biggest on the CoVERT dataset. The reason is that CoVERT is the smallest among the three. A soft prompt-based method can achieve relatively better results with fewer training data and thus, yields the greatest improvement.

As demonstrated in our experiment, a prompt can be considered a means to retrieve information already encoded in the language model — in our case, T5. The data used in the prompt-based approach were converted into the same format utilized to train T5. This helps the language model better understand the intended meaning and purpose of the answers they generate, and in turn improves the quality and relevance of its output.

In addition, the complex nature of COVID-19-related information requires a more nuanced understanding beyond simple keyword matching or hard rules. Soft prompts allow for the incorporation of probabilistic or continuous distributions of words or phrases, capturing the subtleties and context-specific variations present in the text data. This approach enables the model to consider the overall weightage or relevance of different words or phrases, rather than relying on strict binary criteria. In the COVID-19 fact-checking domain, where the relationship between a claim and the evidence can be multifaceted, soft prompts provide the flexibility to account for varying levels of support, refutation, or neutrality. By considering the probabilistic nature of language and the continuous spectrum of textual relationships, soft prompts enable the model to make more accurate and nuanced classifications, enhancing the effectiveness and reliability of COVID-19 fact-checking systems.

The results in Table 4 indicate that soft prompted T5 outperforms vanilla T5 in COVID-19 fact checking, demonstrating the effectiveness of soft prompts. We further conducted experiments to showcase the superiority of soft prompts over hard prompts. To ensure a fair comparison, we manually crafted nine hard prompt templates. The experiment structure is the same as Fig. 3, with the only difference being the use of the following hard prompts in each fact checker, where X represents the text related to COVID-19 to be checked: "Is X a [MASK] statement?", "This is a [MASK] statement: X", "X This is a [MASK] statement", "The statement is [MASK]: X", "X: The statement is [MASK]", "The category of X is [MASK]", "X can be categorized as [MASK]", "[MASK] is the category of X", and "Topic: [MASK] X". In these hard prompts, we varied the location of the [MASK] token in relation to X, the text to be checked, in order to make the structures similar to the soft prompts.

The results of these nine hard prompted T5 classifiers are then ensembled using hard voting to obtain the final results, as shown in Table 5. We can observe that the soft prompts-based ensemble method yields better results. Although well-crafted hard prompts can achieve outstanding results, the process of designing them typically involves carefully crafting specific instructions or cues to guide the model's behavior. This manual definition of hard prompts requires domain expertise or extensive trial and error to identify the most effective prompts. Additionally, there is no widely recognized general guidance or established best practices for designing hard prompts, which can make it challenging to ensure their optimal effectiveness.

On the other hand, soft prompts can be considered as a set of [UNUSED] tokens distributed within the soft prompts template. These [UNUSED] tokens are learned automatically during the model training stage, allowing the model to adapt and assign appropriate weights based on the training data. Soft prompts offer a more automated and adaptable approach, leveraging the model's ability to predict mask tokens and capture relevant contextual information. This eliminates the need for explicit manual crafting of prompts and allows the model to learn from the data itself. Consequently, soft prompting is a more convenient and data-driven approach compared to hard prompting.

In summary, our extensive experiments demonstrate the effectiveness of soft prompts compared to hard prompts and the normal fine-tuning based method.

5.3.3. The impact of the language model on the performance

This paper utilizes T5, an encoder–decoder structure, as the backbone language model. This mechanism differs from popular encoder-based language models, such as BERT, where the encoding of the text is fed to a softmax layer to classify it into predefined labels. In the T5-based model, words are generated and then mapped to the three labels of support, refute, and neutral using cosine distance through the verbalizer, as shown in Section 3.2.1. This structure allows for more expressive and flexible outputs, as the generated words are unconstrained. In this section, we aim to explore the effectiveness of the language model on performance.

We conducted experiments using the same structure as illustrated in Fig. 3, except that BERT is used as the language model. Additionally, the encoding of the inputted text is fed to a softmax layer to classify it into the three predefined labels. As a result, the verbalizer is not needed.

Based on the results in Table 6, it is evident that the T5-based structure outperforms the BERT-based structure. The advantage of T5 lies in its ability to generate a wide range of words and phrases, enabling more nuanced and contextually appropriate responses. This

Table 5
Performance comparison of ensembling hard-prompted and soft-prompted T5.

Prompt type	HealthVer			COVIDFact			CoVERT		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Hard	0.7479	0.7646	0.7623	0.6506	0.720664169	0.683865668	0.6321	0.6005	0.6150
Soft	0.7983	0.7862	0.7901	0.7775	0.7743	0.7759	0.6580	0.6061	0.6265

Table 6
Performance comparison of ensembling T5 and BERT based soft-prompt models.

Language model ensembled	HealthVer			COVIDFact			CoVERT		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
BERT	0.7752	0.7594	0.7647	0.4854	0.7145	0.5781	0.6833	0.5378	0.5625
T5	0.7983	0.7862	0.7901	0.7775	0.7743	0.7759	0.6580	0.6061	0.6265

flexibility is particularly valuable in fact-checking tasks, where the generated outputs must accurately capture the veracity of the analyzed information. By employing a verbalizer, the T5-based structure can map the generated output (i.e., the generated word) of T5 to the desired class labels using cosine distance. On the other hand, BERT solely focuses on encoding input text and mapping it to pre-defined labels (support, refute, and neutral), lacking the flexibility and expressiveness in the label space.

6. Discussion and conclusion

This paper presents a soft prompt-based ensemble learning framework for COVID-19 fact checking. The proposed approach leverages multiple classifiers trained on diverse soft prompts to capture linguistic cues, semantic features, and contextual information in COVID-19-related text data. By combining the collective knowledge and strengths of individual classifiers, the ensemble model achieves robust generalization to new claims. Experimental results demonstrate the effectiveness of prompt-based ensemble learning in improving fact-checking accuracy. The proposed framework contributes to automated fact-checking in the context of COVID-19, empowering users to evaluate online information and combat the dissemination of false or misleading claims.

The proposed method fits well to the scope of a COVID-19 infodemic study. However, it still has limitations. Soft prompts offer limited explicit guidance to the language model, introducing ambiguity and uncertainty in the fact-checking process. The soft prompt-based ensembling method depends greatly on the design of the soft prompt formats, specifically the number of [UNUSED] tokens and their allocation in the design template. Since the underpinning mechanism of the soft prompt is still under investigation, it is hard, if not impossible, to design the most suitable soft prompt for the fact checking task. That is why we ensemble the soft prompts in different formats to leverage the benefit of all the soft prompts and exploit the information on different levels. It is well worth investigating the mechanism of soft prompt so the structure of the proposed method can be simplified. In addition, the current fact checking datasets mainly use three labels – support, refute, and neutral; or two labels – support and refute. However, we think it is necessary to annotate the datasets on a finer level, such as strong support, weak support, neutral, weak refute, and strong refute to quantify the relationship between the claim and evidence in a comprehensive manner. In the current three-label dataset, many weak support or refute data pairs are misclassified as neutral. If the data are annotated with finer labels, this problem can be mitigated.

We plan to improve our method in the following ways in our future work. As mentioned in the limitation, the soft prompt lacks explicit guidance to the language model. This lack of specificity may result in a generation of irrelevant or erroneous responses. Consequently, we aim to investigate the underlying mechanisms of soft prompts to maximize the utility of pretrained language models. Additionally, effective incorporation of domain knowledge, crucial for COVID-19 fact checking, into the language model poses a challenge. Hence, we plan to explore strategies to integrate pertinent knowledge and utilize well-defined prompts tailored to specific applications. Furthermore, our proposed method is presently tailored solely to the English language, limiting its applicability to other languages prevalent in the COVID-19 infodemic. However, developing a new language model from scratch is laborious and resource intensive. Thus, our objective is to fully leverage the existing English model and explore streamlined approaches to efficiently adapt it to other languages. Consequently, our next phase will concentrate on investigating methodologies for language model transferability.

CRediT authorship contribution statement

Shaoqin Huang: Validation, Software, Data curation. **Yue Wang:** Writing – original draft, Supervision, Methodology, Conceptualization. **Eugene Y.C. Wong:** Writing – original draft, Supervision. **Lei Yu:** Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This research is supported by Hong Kong RGC RMGS 700031.

References

Afroz, S., Brennan, M., Greenstadt, R., 2012. Detecting hoaxes, frauds, and deception in writing style online. In: Proceedings of 2012 IEEE Symposium on Security and Privacy. pp. 461–475.
Alhindi, T., Petridis, S., Muresan, S., 2018. Where is your evidence: Improving fact-checking by justification modeling. In: Proceedings of the First Workshop on Fact Extraction and VERification. FEVER, pp. 85–90.
Arana-Catania, M., Kochkina, E., Zubiaga, A., Liakata, M., Procter, R., He, Y., 2022. Natural language inference with self-attention for veracity assessment of pandemic claims. In: Proceedings of NAACL.
Bojanowski, P., Grave, E., Joulin, A., Mikolov, T., 2016. Enriching word vectors with subword information. arXiv preprint arXiv:1607.04606.
Brennen, J.S., Simon, F.M., Howard, P.N., Nielsen, R.K., 2020. Types, Sources, and Claims of COVID-19 Misinformation. Key Findings. Reuters Institute for the Study of Journalism, https://ora.ox.ac.uk/objects/uuid:178db677-fa8b-491d-beda-4bacdc9d7069.
Brown, T.B., Mann, B., et al., 2020. Language models are few-shot learners. In: Proceedings of the 34th International Conference on Neural Information Processing Systems. pp. 1877–1901.
Chen, X., Xie, H., Tao, X., 2022. Vision, status, and research topics of natural language processing. Natural Lang. Process. J. 1, 100001.
Ciampaglia, G.L., Shiralkar, P., Rocha, L.M., Bollen, J., Menczer, F., Flammini, A., 2015. Computational fact checking from knowledge networks. PLoS One 10 (6), 1–13.
Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of NAACL-HLT. pp. 4171–4186.

Dong, X., Yu, Z., Cao, W., Shi, Y., Ma, Q., 2020. A survey on ensemble learning. Front. Comput. Sci. 14, 241–258.
Enos, F., Shriberg, E., Graciarena, M., Hirschberg, J., Stolcke, A., 2007. Detecting deception using critical segments. In: Proceedings of International Speech Communication Association. pp. 2281–2284.
Feng, S., Banerjee, R., Choi, Y., 2012. Syntactic stylometry for deception detection. In: Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics. pp. 171–175.
Ferreira, W., Vlachos, A., 2016. Emergent: a novel data-set for stance classification. In: Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 1163–1168.
Guo, Z., Schlichtkrull, M., Vlachos, A., 2022. A survey on automated fact-checking. Trans. Assoc. Comput. Linguist. 10, 178–206.
Hanselowski, A., Stab, C., Schulz, C., Li, Z., Gurevych, I., 2019. A richly annotated corpus for different tasks in automated fact-checking. In: Proceedings of the 23rd Conference on Computational Natural Language Learning. CoNLL, pp. 493–503.
Khan, W., Daud, A., Khan, K., Muhammad, S., Haq, R., 2023. Exploring the frontiers of deep learning and natural language processing: A comprehensive overview of key challenges and emerging trends. Natural Lang. Process. J. 4, 100026.
Kotonya, N., Toni, F., 2020. Explainable automated fact-checking: A survey. In: Proceedings of the 28th International Conference on Computational Linguistics. pp. 5430–5443.
Lee, N., Li, B.Z., Wang, S., Yih, W.T., Ma, H., Khabsa, M., 2020. Language models as fact checkers? In: Proceedings of the Third Workshop on Fact Extraction and VERification. FEVER, pp. 36–41.
Li, S., Hu, X., Lin, L., Wen, L., 2022. Pair-level supervised contrastive learning for natural language inference. In: Proceedings of 2022 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 8237–8241.
Li, X.L., Liang, P., 2021. Prefix-tuning: Optimizing continuous prompts for generation. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics. pp. 4582–4597.
Lin, Y.C., Su, K.Y., 2021. How fast can BERT learn simple natural language inference? In: Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics. pp. 626–633.
Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., Neubig, G., 2023. Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. ACM Comput. Surv. 55 (9), 195.
Luo, L., Wang, Y., 2019. Emotionx-HSU: Adopting pre-trained BERT for emotion classification. arXiv preprint arXiv:1907.09669.
Luo, L., Wang, Y., Liu, H., 2022. COVID-19 personal health mention detection from tweets using dual convolutional neural network. Expert Syst. Appl. 200, 117139.
Luo, L., Wang, Y., Mo, D.Y., 2023. Identifying heart disease risk factors from electronic health records using an ensemble of deep learning method. IISE Trans. Healthc. Syst. Eng. 13 (3), 237–247.
Mienye, I.D., Sun, Y., 2022. A survey of ensemble learning: Concepts, algorithms, applications, and prospects. IEEE Access 10, 99129–99149.
Mihalcea, R., Strapparava, C., 2009. The lie detector: Explorations in the automatic recognition of deceptive language. In: Proceedings of the ACL-IJCNLP 2009 Conference. pp. 309–312.
Mohr, I., Wührl, A., Klinger, R., 2022. CoVert: A corpus of fact-checked biomedical COVID-19 tweets. In: Proceedings of the 13th Conference on Language Resources and Evaluation. LREC 2022, pp. 244–257.
Ott, M., Choi, Y., Cardie, C., Hancock, J.T., 2011. Finding deceptive opinion spam by any stretch of the imagination. In: Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics. pp. 309–319.
Peng, Z., Li, M., Wang, Y., Ho, G., 2023. Combating the COVID-19 infodemic using prompt-based curriculum learning. Expert Syst. Appl. (accepted).
Popat, K., Mukherjee, S., Yates, A., Weikum, G., 2018. Declare: Debunking fake news and false claims using evidence-aware deep learning. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. pp. 22–32.
Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., Liu, P.J., 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. J. Mach. Learn. Res. 21 (1), 140.
Ross, A., Pavlick, E., 2019. How well do NLI models capture verb veridicality? In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing. EMNLP-IJCNLP, pp. 2230–2240.
Saakyan, A., Chakrabarty, T., Muresan, S., 2021. COVID-fact: Fact extraction and verification of real-world claims on COVID-19 pandemic. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 730–742.
Sarrouti, M., Abacha, A.B., Mrabet, Y., Demner-Fushman, D., 2021. Evidence-based fact-checking of health-related claims. In: Findings of the Association for Computational Linguistics: EMNLP 2021. pp. 3499–3512.
Schick, Y., Schütze, H., 2021. Exploiting cloze questions for few-shot text classification and natural language inference. In: Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics. pp. 255–269.
Sung, M., Lee, J., Yi, S., Jeon, M., Kim, S., Kang, J., 2021. Can language models be biomedical knowledge bases? In: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing. pp. 4723–4734.
Sushil, M., Suster, S., Daelemans, W., 2021. Are we there yet? Exploring clinical domain knowledge of BERT models. In: Proceedings of the 20th Workshop on Biomedical Language Processing. pp. 41–53.
Talman, A., Chatzikyriakidis, S., 2019. Testing the generalization power of neural network models across NLI benchmarks. In: Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP. pp. 85–94.
Vlachos, A., Riedel, S., 2014. Fact checking: Task definition and dataset construction. In: Proceedings of the ACL 2014 Workshop on Language Technologies and Computational Social Science. pp. 18–22.
Vlachos, A., Riedel, S., 2015. Identification and verification of simple claims about statistical properties. In: Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing. pp. 2596–2601.
Wadden, D., Lin, S., Lo, K., Wang, L.L., Zuylen, M.van., Cohan, A., Hajishirzi, H., 2020. Fact or fiction: Verifying scientific claims. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing. EMNLP, pp. 7534–7550.
Wadden, D., Lo, K., Wang, L.L., Cohan, A., Hajishirzi, H., 2021. Longchecker: Improving scientific claim verification by modeling full-abstract context. arXiv preprint. arxiv:2112.01640.
Wang, Y., Li, X., Mo, D.Y., 2021. Knowledge-empowered multitask learning to address the semantic gap between customer needs and design specifications. IEEE Trans. Ind. Inform. 17 (12), 8397–8405.
Wang, Y., Li, X., Tsung, F., 2020a. Configuration-based smart customization service: A multitask learning approach. IEEE Trans. Autom. Sci. Eng. 17 (4), 2038–2047.
Wang, Y., Li, X., Zhang, L.L., Mo, D.Y., 2022. Configuring products with natural language: a simple yet effective approach based on text embeddings and multilayer perceptron. Int. J. Prod. Res. 60 (17), 5394–5406.
Wang, L.L., Lo, K., Chandrasekhar, Y., et al., 2020b. CORD-19: The COVID-19 open research dataset. In: Proceedings of the 1st Workshop on NLP for COVID-19.
Wührl, A., Klinger, R., 2022. Entity-based claim representation improves fact-checking of medical content in tweets. In: Proceedings of the 9th Workshop on Argument Mining. pp. 187–198.
Zhuang, Z., Liang, Z., Rao, Y., Xie, H., Wang, F.L., 2023. Out-of-vocabulary word embedding learning based on reading comprehension mechanism. Natural Lang. Process. J. 5, 100038.